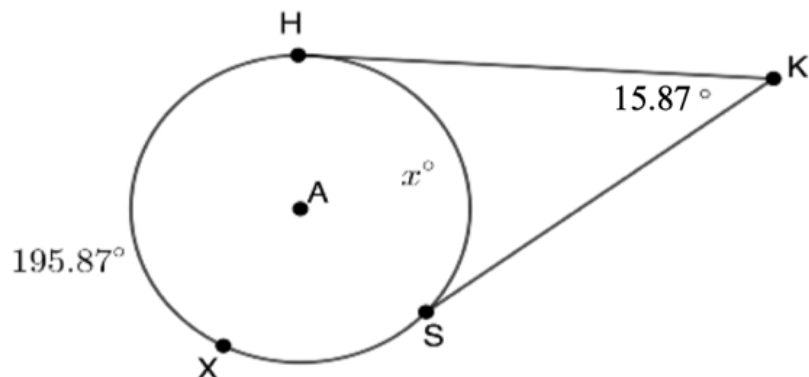


Geometry
Unit 10
Lesson 9 Practice

Name _____ **Answer Key** _____
Date _____
Period _____

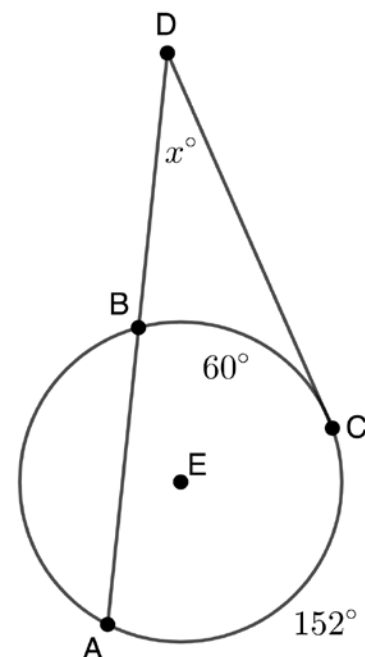
For questions 1-2, Circle A has points H, X , and S on the circle. Tangents $\overline{HK}$ and $\overline{SK}$ intersect at point K outside the circle. The $m\widehat{HXS} = 195.87^\circ$, $m\widehat{HS} = x^\circ$, and $m\angle HKS = 15.87^\circ$.



- Write an equation that can be used to find the value of x .
 $360 - 195.87 = x$ or $\frac{195.87 - x}{2} = 15.87$

- Find the value of x .
 $x = 360 - 195.87 = 164.13$

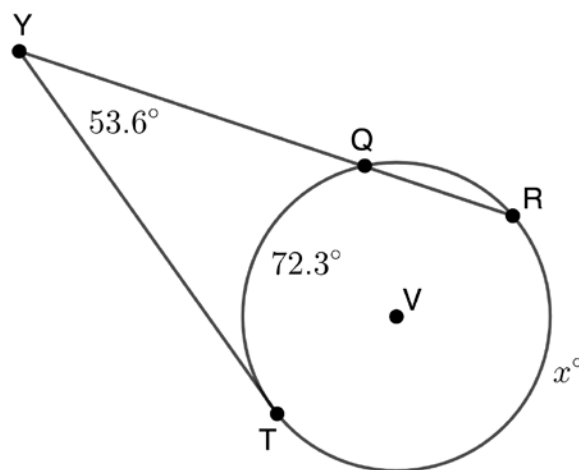
For questions 3-4, Circle E has points A, B , and C on the circle. Tangent $\overline{CD}$ and secant $\overline{AD}$ intersect at point D outside the circle. The $m\widehat{AC} = 152^\circ$, $m\widehat{BC} = 60^\circ$, and $m\angle BDC = x^\circ$.



- Write an equation that can be used to find the value of x .
 $x = \frac{152 - 60}{2}$

- Find the value of x .
 $x = \frac{152 - 60}{2} = 46$

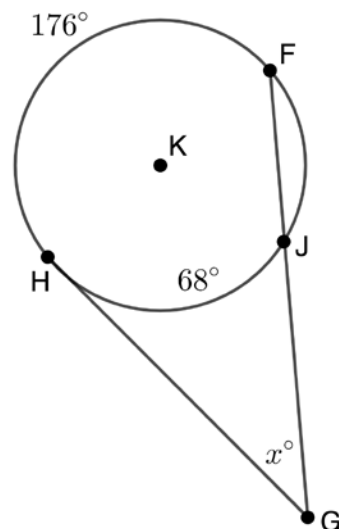
For questions 5-6, Circle V has points T, Q , and R on the circle. Tangent $\overline{TY}$ and secant $\overline{RY}$ intersect at point Y outside the circle. The $m\widehat{TR} = x^\circ$, $m\widehat{TQ} = 72.3^\circ$, and $m\angle TYQ = 53.6^\circ$.



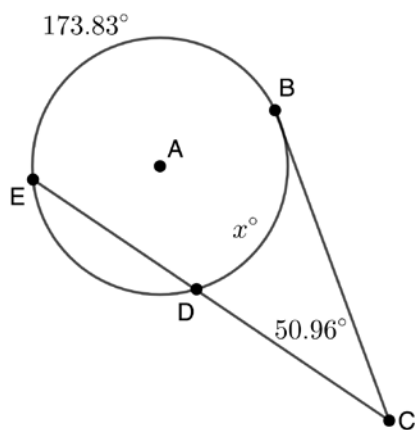
- Write an equation that can be used to find the value of x .
 $\frac{x - 72.3}{2} = 53.6$

- Find the value of x .
 $\frac{x - 72.3}{2} = 53.6$
 $x - 72.3 = 107.2$
 $x = 179.5$

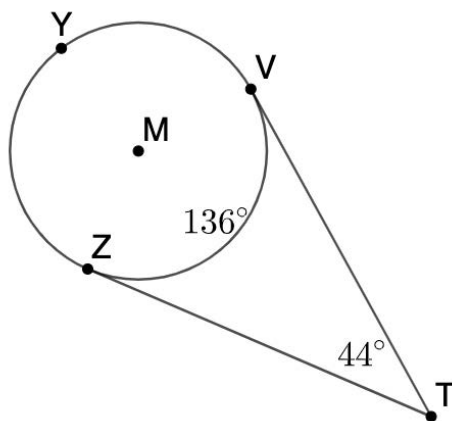
7. Circle K has points F, H , and J on the circle. Tangent $\overline{HG}$ and secant $\overline{FG}$ intersect at point G outside the circle. The $m\widehat{HF} = 176^\circ$, $m\widehat{HJ} = 68^\circ$, and $m\angle JGH = x^\circ$. What is the $m\angle JGH$?
- $$m\angle JGH = \frac{176 - 68}{2} = 54^\circ$$



8. Circle A has points B, D , and E on the circle. Tangent $\overline{BC}$ and secant $\overline{EC}$ intersect at point C outside the circle. The $m\widehat{EB} = 173.83^\circ$, $m\widehat{DB} = x^\circ$, and $m\angle BCD = 50.96^\circ$. What is the value of x ?
- $$\frac{173.83 - x}{2} = 50.96$$
- $$173.83 - x = 101.92$$
- $$x = 71.91$$



9. Circle M has points V, Y , and Z on the circle. Tangents $\overline{ZT}$ and $\overline{VT}$ intersect at point T outside the circle. $m\widehat{VZ} = 136^\circ$, and $m\angle VTZ = 44^\circ$. What is the $m\widehat{VYZ}$?
- $$\frac{x - 136}{2} = 44$$
- $$x - 136 = 88$$
- $$x = 224$$
- $$m\widehat{VYZ} = 224^\circ$$



10. Circle O has points K, M , and N on the circle. Tangents $\overline{KL}$ and $\overline{ML}$ intersect at point L outside the circle. The $m\widehat{MNK} = 227^\circ$, $m\widehat{MK} = 133^\circ$, and $m\angle KLM = x^\circ$. What is the $m\angle KLM$?
- $$m\angle KLM = \frac{227 - 133}{2} = 47^\circ$$

